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K J SOMAIYA INSTITUTE OF TECHNOLOGY, SION, MUMBAI

DEPARTMENT OF COMPUTER ENGINEERING

AgriSense: AI-Driven Crop Type, Phenology, Moisture Stress and Irrigation Advisory Platform for Karnal

TY Minor Project Report

**Submitted in partial fulfillment of the requirements
of the Subject Project Based Learning: Minor Project Lab-I (Sem V)**

by

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CERTIFICATE



*This is to certify that the project entitled “**AgriSense: AI-Driven Crop Type, Phenology, Moisture Stress and Irrigation Advisory Platform for Karnal**” is bonafide work of **Atharva Naik, Rahul Meshram, Daksh Patel** submitted as a TY Sem V Minor project in the subject of Project Based Learning: Minor Project Lab-I, Computer Engineering for the academic year 2026-27.*

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PROJECT APPROVAL FOR T. Y.

This project report entitled “AgriSense: AI-Driven Crop Type, Phenology, Moisture Stress and Irrigation Advisory Platform for Karnal”

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DECLARATION

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea, data, fact, source, model result or experiment in our submission. Dataset-derived observations, prototype model outputs and independently validated measurements are explicitly distinguished in this report. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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ABSTRACT

AgriSense is a remote-sensing and artificial-intelligence-based decision-support platform developed for crop monitoring and irrigation advisory in Karnal District, Haryana. The system integrates Sentinel-2 optical data, Sentinel-1 SAR observations, weather information, soil properties, machine-learning models, and a FAO-56-based crop water-balance approach within a role-specific web application.

The current implementation uses satellite observations from the Kharif 2025 season sampled at 4,311 fixed locations, producing 43,110 temporal records. Optical observations were affected by cloud cover, while Sentinel-1 SAR maintained complete coverage across all satellite periods. This motivated the use of optical–radar data fusion and independent tracking of valid and carried-forward observations. Weather variables and soil characteristics are aligned through a 19-step, 8-day seasonal timeline to provide continuous monitoring without generating artificial satellite observations.

The platform includes separate engines for crop classification, phenological-stage estimation, moisture-stress detection, and irrigation advisory. Random Forest, XGBoost, SVM, Logistic Regression, and LSTM models are evaluated depending on the task. The application provides seasonal monitoring, model inference, spatial map layers, and role-specific interfaces for farmers, agriculture officers, irrigation authorities, GIS analysts, and researchers.

The current evaluation is considered prototype-level because some crop, stress, and phenology targets are derived or pseudo labels. The work therefore demonstrates a reproducible multi-sensor architecture and establishes a foundation for future field-based validation and operational deployment.

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LIST OF ABBREVIATIONS

Sr. No.	Abbreviation	Description
1	AI	Artificial Intelligence
2	API	Application Programming Interface
3	ARD	Analysis Ready Data
4	CM	Confusion Matrix
5	ET ₀	Reference Evapotranspiration
6	ET _c	Crop Evapotranspiration
7	EVI	Enhanced Vegetation Index
8	FAO	Food and Agriculture Organization
9	GEE	Google Earth Engine
10	GIS	Geographic Information System
11	K _c	Crop Coefficient
12	LSTM	Long Short-Term Memory
13	ML	Machine Learning
14	NDMI	Normalized Difference Moisture Index
15	NDVI	Normalized Difference Vegetation Index
16	NDWI	Normalized Difference Water Index
17	RF	Random Forest
18	SAR	Synthetic Aperture Radar
19	SAVI	Soil Adjusted Vegetation Index
20	SVM	Support Vector Machine
21	UI	User Interface
22	UTM	Universal Transverse Mercator
23	WUA	Water User Association
24	XGB	XGBoost

CHAPTER 1

INTRODUCTION

1.1 Problem Definition

Agriculture in Karnal requires timely information about crop type, crop growth stage, moisture stress, and irrigation requirements. Although field surveys provide reliable observations, carrying them out repeatedly over large agricultural areas is time-consuming and resource-intensive. Satellite remote sensing provides a practical alternative for monitoring crops over large regions, but no single satellite sensor can capture all the required information under every condition.

Sentinel-2 provides multispectral observations that are useful for analysing vegetation condition and moisture-related changes. However, during the Kharif monsoon season, frequent cloud cover can significantly reduce the availability of usable optical imagery. Sentinel-1 Synthetic Aperture Radar (SAR), on the other hand, operates independently of cloud conditions and provides VV and VH backscatter measurements that can complement optical observations.

To address these limitations, AgriSense combines optical, SAR, weather, and soil information rather than depending on a single source. The current Karnal dataset contains observations from 4,311 fixed locations across ten satellite periods, resulting in 43,110 temporal records. Sentinel-1 observations are available for all ten periods, whereas Sentinel-2 coverage varies considerably and is completely unavailable during some periods because of cloud contamination.

An important requirement of the system is therefore to handle missing observations transparently. AgriSense keeps track of whether an observation is newly available or carried forward from an earlier period and records the age of carried observations. Missing satellite images are not artificially generated simply to maintain a fixed timeline. This allows the system to preserve the actual availability and limitations of the remote-sensing data.

1.2 Aim and Objectives

Aim:

To design and implement AgriSense, an integrated satellite remote-sensing and AI-based decision-support platform for Karnal District that supports crop identification, phenological-stage monitoring, moisture-stress detection, and 8-day irrigation advisory using optical, microwave, weather, and soil information.

Objectives:

- 1.To develop a reproducible processing pipeline for Sentinel-2 optical indices and Sentinel-1 VV/VH SAR observations over Karnal District.
- 2.To construct a temporal feature dataset while preserving the actual availability and validity of optical and SAR observations.
- 3.To evaluate Random Forest, XGBoost, and SVM models for crop-type classification.
- 4.To compare Random Forest and XGBoost models with an LSTM-based temporal model for crop phenology estimation.
- 5.To evaluate Logistic Regression, Random Forest, and XGBoost models for moisture-stress classification.
- 6.To integrate 19 real 8-day weather periods and a 24-band soil dataset with the satellite observation timeline.

- 7.To implement an FAO-56-based crop water-balance model together with irrigation-priority rules.
- 8.To develop a role-based web application that provides seasonal monitoring, map visualization, field-level inspection, model evaluation, and persistent Python-based inference.

1.3 Organization of the Report

The remainder of this report is organized as follows:

Chapter	Content Description
Chapter 2	Reviews previous research related to crop classification, optical-SAR fusion, crop phenology, moisture-stress monitoring, evapotranspiration estimation, and irrigation scheduling.
Chapter 3	Describes the functional and non-functional requirements of AgriSense along with the hardware and software requirements of the prototype.
Chapter 4	Presents the overall system architecture, workflow, UML diagrams, including use case, activity, sequence and class diagrams, and the project development timeline.
Chapter 5	Explains data acquisition, temporal alignment, feature engineering, machine-learning models, LSTM sequence preparation, crop water-balance calculations, and API and data structures.
Chapter 6	Describes the implementation of the Karnal dataset pipeline and web application, live inference architecture, satellite observation coverage, and prototype model results.
Chapter 7	Summarizes the outcomes of the project, discusses current limitations, and presents the proposed roadmap for validation and future deployment.
Chapter 8	Lists the scientific, methodological, and technical sources referred to throughout the report.

Throughout the project, a clear distinction is maintained between directly measured observations, variables derived through processing, and outputs generated by the AI models. This is important because the current crop, phenology, and moisture-stress labels are primarily derived or pseudo labels and have not yet been fully validated using independent field-survey ground truth. The current implementation should therefore be considered a prototype framework that demonstrates the complete data-processing, modelling, and decision-support workflow of AgriSense.

CHAPTER 2

REVIEW OF LITERATURE

2.1 Introduction

Remote-sensing-based agricultural monitoring has developed from single-date crop mapping to multi-temporal and multi-sensor analysis. Sentinel-2 provides multispectral information suitable for vegetation and moisture-sensitive indices, while Sentinel-1 SAR provides cloud-independent VV and VH observations. Recent studies increasingly combine these datasets with machine learning, phenological information, weather data, and cloud-based processing platforms such as Google Earth Engine.

These techniques are particularly relevant to Karnal because the Kharif season is affected by frequent monsoon cloud cover. Although Sentinel-2 provides high-resolution multispectral observations, cloud contamination can create major gaps in the optical time series. Sentinel-1 therefore provides an important complementary source for continuous seasonal monitoring.

This review focuses on research related to crop classification, optical-SAR fusion, crop phenology, moisture-stress detection, evapotranspiration, irrigation scheduling, and scalable geospatial processing.

2.2 Literature Survey

Crop Classification and Sentinel-1/Sentinel-2 Fusion

Sentinel-2 provides multispectral surface reflectance at spatial resolutions suitable for field-scale crop monitoring, while Sentinel-1 C-band SAR provides all-weather backscatter measurements that are especially valuable during cloudy monsoon periods [1][2]. Studies combining optical and SAR time series show that complementary spectral and structural information can improve crop discrimination compared with a single sensor [7]. This directly motivates the AgriSense decision to retain both Sentinel-2-derived indices and Sentinel-1 VV/VH features. Mishra et al. studied rainy-season crop classification near Karnal using Sentinel-1 C-band SAR time-series data and dual-polarization vegetation indices. Random Forest and SVM achieved accuracies of 93.77% and 93.50%, respectively [5]. The study demonstrates the usefulness of SAR for crop monitoring in cloudy conditions.

Ahlawat et al. used multi-date Sentinel-1 observations for regional crop mapping in Hisar District, Haryana [6]. The study successfully distinguished crops such as wheat, mustard, and sugarcane using their temporal radar behaviour, showing the potential of repeated SAR observations for agricultural mapping in Haryana.

Cai et al. combined Sentinel-1, Sentinel-2, NDVI, phenological parameters, and Random Forest for paddy rice mapping [7]. Their approach achieved an overall accuracy above 95%, highlighting the benefit of combining temporal optical and radar information.

Similarly, Blickensdörfer et al. combined Sentinel-1, Sentinel-2, Landsat, and environmental information for large-scale crop classification [8]. Their results showed that multi-source temporal features can improve the robustness of crop mapping.

These studies support the use of optical-SAR fusion in AgriSense, particularly where optical observations may be unavailable because of cloud cover.

Crop Phenology from Optical and SAR Time Series

Woźniak et al. developed phenological indices using Sentinel-1 time-series observations for crop classification [9]. Their approach achieved regional overall accuracies of approximately 86–90%, indicating that seasonal crop development can also be observed through radar backscatter.

Meroni et al. compared Sentinel-1 and Sentinel-2 observations for crop land-surface phenology [10]. Their results showed that radar and optical measurements provide complementary information, although their agreement varies depending on crop type and growth stage.

Liao et al. used Sentinel-2 time-series information for near-real-time crop phenology detection [11]. Their work demonstrated that growth-stage estimation benefits from analysing temporal patterns rather than treating each satellite observation independently.

These findings motivate the use of sequential features in AgriSense, where optical indices and SAR backscatter are analysed together through both conventional machine-learning models and an LSTM-based temporal model.

Moisture Stress and Irrigation Scheduling

Crop moisture stress cannot always be identified reliably using a single vegetation index because the observed signal is influenced by crop stage, canopy structure, soil background, weather, and management practices.

Veysi et al. evaluated a satellite-based crop water stress index for irrigation scheduling and compared the satellite results with field observations of canopy temperature and vegetation water content [13]. Their work demonstrates the potential of remote sensing for irrigation support while also emphasizing the importance of field validation.

AgriSense combines optical moisture indicators such as NDMI and NDWI with SAR measurements and phenological information to estimate moisture stress. However, because the current stress labels are derived or pseudo labels, the model results are considered prototype-level rather than independently validated agronomic measurements.

For irrigation estimation, AgriSense follows the FAO-56 crop evapotranspiration framework developed by Allen et al. [3]. Reference evapotranspiration is calculated from meteorological conditions, while crop evapotranspiration is estimated as:

$$ET_c = K_c \times ET_0$$

where ET_0 is reference evapotranspiration and K_c represents the crop coefficient according to crop type and growth stage. Effective rainfall and estimated crop-water requirements can then be used to determine irrigation priority.

Weather information can be obtained from sources such as ERA5-Land, which provides meteorological and land-surface variables suitable for environmental and hydrological applications [2]. In AgriSense, weather information is organized into 8-day periods so that irrigation advisories can continue even when new satellite observations are unavailable.

Machine Learning and Geospatial Processing

Random Forest is widely used for crop classification because it can handle nonlinear relationships and mixed remote-sensing features while remaining suitable for relatively small datasets [15]. XGBoost provides gradient-boosted decision trees capable of modelling more complex feature interactions [16]. Support Vector Machines have also shown strong performance in remote-sensing classification.

For temporal modelling, Long Short-Term Memory networks are designed to learn relationships across sequential observations [17]. This makes LSTM suitable for phenology analysis, where changes in vegetation indices and SAR backscatter across multiple dates are important.

Google Earth Engine is widely used for satellite-data preprocessing because it combines large remote-sensing archives with cloud-based computation [4]. AgriSense uses GEE for preparing Sentinel-1, Sentinel-2, weather, and soil datasets, while downloaded GeoTIFF and CSV files are used for local modelling and application development.

Table 2.1: Summary of Literature Survey - Related Works

Study	Main Focus	Method/ Data	Key Findings
Mishra et al. [5]	Kharif crop classification near Karnal	Sentinel-1 VV/VH, RF, SVM	RF: 93.77%; SVM: 93.50%
Ahlawat et al. [6]	Haryana crop mapping	Multi-date Sentinel-1	Demonstrated regional SAR-based crop mapping
Cai et al. [7]	Paddy rice mapping	Sentinel-1 + Sentinel-2 + RF	Overall accuracy above 95%
Blickensdörfer et al. [8]	Large-scale crop mapping	S1 + S2 + Landsat + RF	Multi-source data improved robustness
Woźniak et al. [9]	Crop phenology	Sentinel-1 time series	Approximately 86–90% regional accuracy
Meroni et al. [10]	Land-surface phenology	Sentinel-1 + Sentinel-2	Optical and SAR phenology were complementary
Liao et al. [11]	Growth-stage estimation	Sentinel-2 time series	Temporal information supported detailed phenology estimation
Veysi et al. [13]	Water stress and irrigation	Satellite CWSI + field data	Demonstrated satellite-based stress monitoring
Allen et al. [3]	Crop water requirement	FAO-56 Penman-Monteith	Established standard ET_0 and Kc framework

2.2 Limitations of Existing Systems

The reviewed studies demonstrate that satellite time series, machine learning, and evapotranspiration models can support crop monitoring and irrigation management. However,

several limitations remain when these methods are integrated into an operational agricultural decision-support system.

Table 2.2: Limitations Identified in Existing Systems

Limitation	Effect	AgriSense Approach
Optical cloud dependence	Monsoon clouds can remove important Sentinel-2 observations	Sentinel-1 is maintained as an independent all-weather data source
Dependence on ground truth	High accuracy often requires extensive field labels	Current results are treated as prototype results until field validation
Phenology interpretation	Remote-sensing phenology may not directly represent agronomic stages	Temporal modelling is used, with future validation using observed stage labels
Single-task systems	Many studies address only crop mapping, stress, or irrigation	AgriSense combines crop, phenology, stress, weather, soil, and irrigation analysis
Temporal mismatch	Different datasets have different observation frequencies	An 8-day master timeline aligns the available data while preserving original observation dates
Irrigation validation	Estimated stress may not correspond directly to actual irrigation requirements	Future validation will use soil-moisture and irrigation records
Regional transferability	Models developed in one region or season may perform differently elsewhere	Spatially grouped and multi-season validation is planned

The main gap identified from the literature is therefore not the absence of individual crop-classification or irrigation methods, but the limited integration of these methods into a single seasonal decision-support system. AgriSense addresses this gap by combining optical and SAR observations with weather, soil information, machine learning, temporal modelling, and irrigation estimation while preserving information about data availability and validation status.

The current system should therefore be viewed as an integrated prototype that establishes the complete processing and decision-support workflow, with independent field-based validation forming an important part of future development.

CHAPTER 3

REQUIREMENT SPECIFICATION

3.1 Introduction

AgriSense is implemented as a data-intensive web application with an offline geospatial preprocessing layer, machine-learning artifacts, a persistent Python inference service, an Express web/API layer and a browser-based interactive map. The requirements below represent the current Karnal prototype rather than a hypothetical large-scale production deployment.

3.2 Hardware Requirements

Table 3.1: Hardware Requirements Specification

Component	Specification	Purpose
Development CPU	4+ modern CPU cores	Raster sampling, JSON generation and local API services.
Development RAM	8 GB minimum; 16 GB recommended	Handles 43,110-row temporal table, raster access and model artifacts.
Storage	20 GB free SSD minimum	Old satellite tiles, weather/soil rasters, temporal tables and artifacts.
Training Compute	Google Colab CPU/GPU as available	Model training, especially LSTM phenology experiments.
Client Device	Modern browser-capable laptop/desktop	Leaflet map, Chart.js analytics and field inspector.
Network	Stable internet for initial data/API setup	GEE/Colab/GitHub access and optional deployment.

3.3 Software Requirements

Table 3.2: Software Requirements - Technology Stack

Layer	Technology	Role
Geospatial Cloud	Google Earth Engine	Earth observation preprocessing and export.
Language	Python 3.x	Raster sampling, temporal fusion, ML preprocessing and FastAPI inference.
Raster Processing	Rasterio / PyProj	GeoTIFF access, CRS transformation and point sampling.
Data Processing	Pandas / NumPy	Temporal tables, QA summaries and master-period generation.
ML - Classical	scikit-learn	Random Forest, SVM, Logistic Regression, preprocessing and metrics.
ML - Boosting	XGBoost	Boosted tree models for comparison and stress deployment.
ML - Temporal	TensorFlow / Keras	Phenology LSTM training and serialized inference.
Inference API	FastAPI + Uvicorn	Loads selected artifacts once and returns predictions.

Layer	Technology	Role
Web/API Server	Node.js + Express	Static app hosting, data endpoints and inference proxy.
Mapping	Leaflet + Esri imagery	Interactive Karnal map and observation-point visualization.
Charts	Chart.js	Temporal profiles, QA distributions and model evaluation views.
Version Control	Git / GitHub	Source control and project repository management.

External Data Dependencies:

- Sentinel-2 surface reflectance and Sentinel-1 GRD SAR from the Copernicus programme [1][2].
- CHIRPS precipitation for eight-day rainfall aggregation [3].
- ERA5-Land meteorological variables for temperature, dew point, wind, pressure, radiation and potential evaporation [4].
- OpenLandMap-style static soil properties at six standard depths for texture, bulk density, organic carbon and field capacity [5].
- Karnal district boundary and the project-specific observation-point CSV used to sample the old satellite tiles.

CHAPTER 4

PROJECT ANALYSIS & DESIGN

4.1 Introduction

The design evolved from a single GEE pipeline into a hybrid architecture because the new eight-day satellite export could not be downloaded completely under the available non-commercial processing quota. The final design therefore preserves the real old satellite dataset, supplements it with the complete new weather and soil products, and makes the temporal mismatch explicit in both processing and user interface.

4.2 System Architecture

The architecture separates data acquisition, local temporal processing, AI inference, water-balance logic and presentation. Ten old satellite dates are converted to a fixed-location temporal table. Nineteen eight-day weather periods and the static soil stack are sampled to the same observation locations. A master timeline then decides whether optical and SAR inputs are fresh, carried or unavailable at each eight-day step. The selected models are served through FastAPI and proxied through the Express server to the browser.

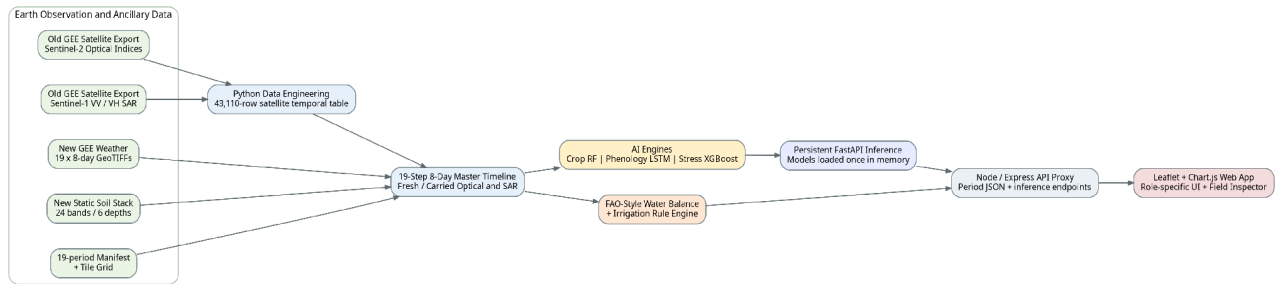


Figure 4.1: System Architecture Diagram - Hybrid Data and Inference Architecture

4.3 System Flow Diagram

The application flow begins with role selection and period selection. The chosen master period loads the real eight-day weather context and the latest independently valid optical and SAR states. Users then choose the information layer permitted by their role. Field selection can trigger the trained Crop RF, Phenology LSTM and Stress XGBoost inference services as well as the eight-day water-balance calculation. The interface returns the result together with source dates and provenance labels.

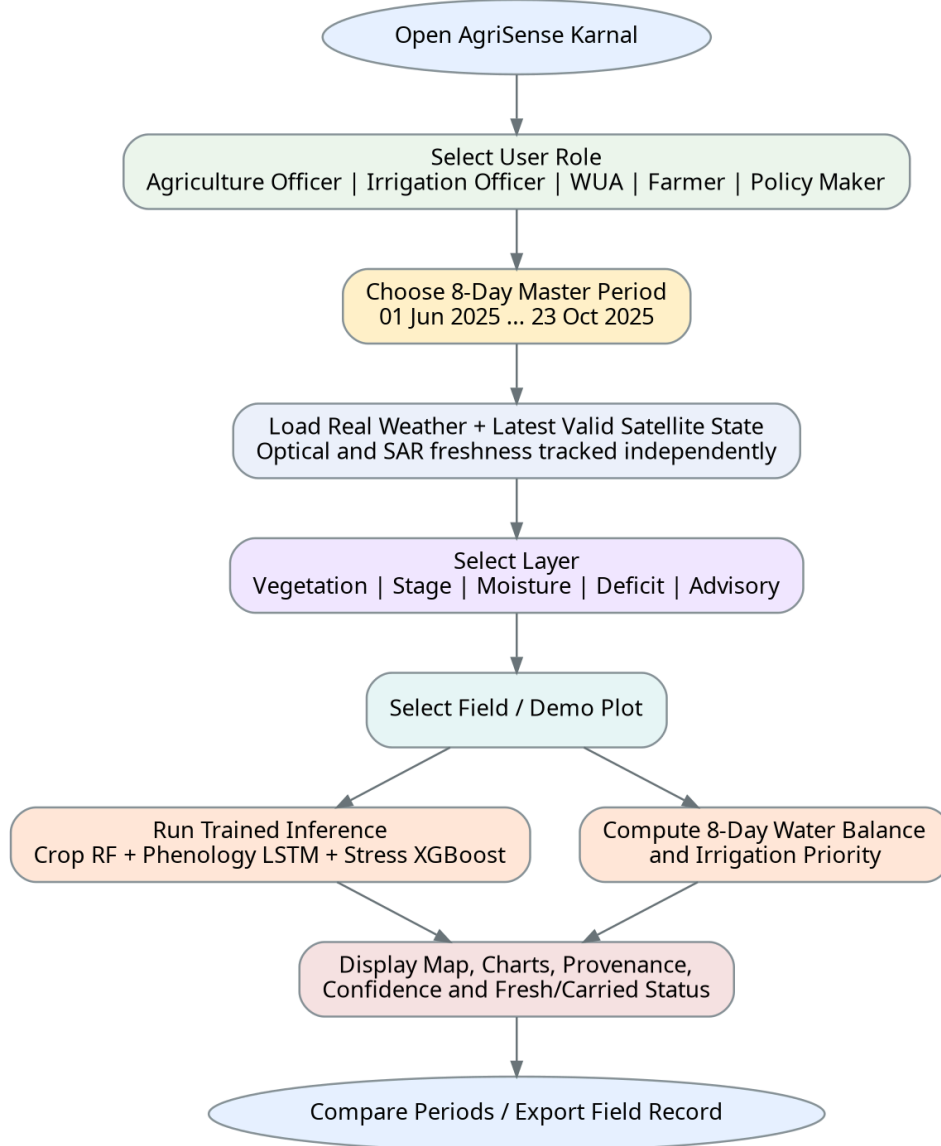


Figure 4.2: System Flow Diagram

4.4 Timeline Chart

Table 4.1: Project Development Timeline (Indicative Gantt Summary)

Phase	Activity	Duration	Week No.
Phase 1	Problem definition, literature and research questions	1 week	Week 1
Phase 2	Old GEE satellite extraction and initial demo	2 weeks	Weeks 2-3
Phase 3	New GEE weather/soil/manifest export	2 weeks	Weeks 3-4
Phase 4	Real temporal-table construction and QA	2 weeks	Weeks 5-6
Phase 5	Crop, phenology and stress model experiments	3 weeks	Weeks 6-8
Phase 6	19-step eight-day master-timeline application	2 weeks	Weeks 8-9
Phase 7	Role-specific UI and persistent inference integration	2 weeks	Weeks 9-10

Phase	Activity	Duration	Week No.
Phase 8	Testing, result consolidation and documentation	2 weeks	Weeks 11-12

Project Development Timeline (Indicative)

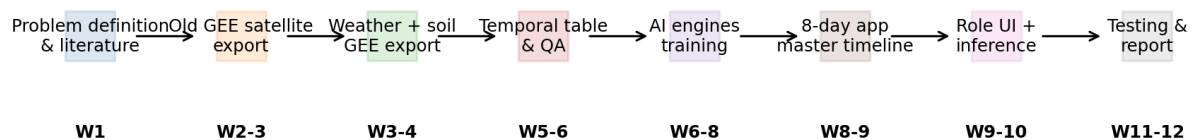


Figure 4.3: Project Development Timeline

4.5 Module Interaction Diagram

The module interaction separates the user-facing role logic from data and model services. Period JSON files provide the current eight-day state, the real temporal CSV provides the sequence history required by the LSTM, private serialized model artifacts are loaded only by the Python inference service, and the Express layer acts as the single browser-facing API entry point. This avoids exposing .joblib or .keras files directly through the public web server.

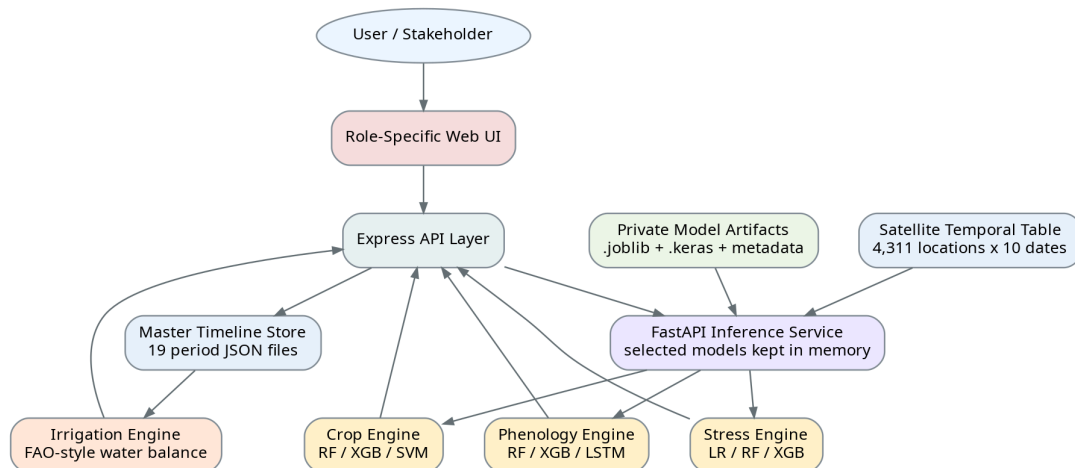


Figure 4.4: Module Interaction Diagram - Data Flow Between Components

CHAPTER 5

METHODOLOGY

5.1 Introduction

This chapter describes the end-to-end methodology implemented for the Karnal prototype. Particular emphasis is placed on preserving real temporal availability, correcting sensor-specific features, preventing leakage during machine-learning evaluation and keeping the irrigation logic transparent.

5.2 Project Module Details

Module 1: Hybrid Data Acquisition

The real satellite component originates from the first successful GEE export. The local folder contains 110 files across ten dates. Raster preflight identified 88 full 14-band tiles and 22 three-band SAR-only tiles. The supported old-script layouts are therefore 14 bands for optical plus SAR and 3 bands for SAR-only periods. The 14-band layout contains B2, B3, B4, B8, B11, B12, NDVI, NDWI, NDMI, EVI, SAVI, VV, VH and the old raw ratio. The raw ratio is retained only for traceability; the modelling feature is recomputed as $VV_VH_ratio_dB = VV - VH$ because Sentinel-1 GRD values are already in dB.

The new GEE workflow successfully downloaded nineteen eight-day weather rasters, one static 24-band soil stack, the period manifest and the 24-tile export grid. Every weather raster contains rainfall, mean/max/min temperature, dew point, 10 m wind speed, surface pressure, solar radiation, ERA5 potential evaporation and a weather-valid flag. The soil stack contains texture, bulk density, organic carbon and field capacity at 0, 10, 30, 60, 100 and 200 cm depths.

Table 5.1: Karnal Dataset Components Used by AgriSense

Component	Size / Frequency	Use
Old satellite GeoTIFFs	110 files / 10 dates	Sentinel-2-derived indices + Sentinel-1 SAR; 14-band and SAR-only 3-band layouts.
Real satellite temporal table	43,110 rows	4,311 fixed Karnal locations x 10 dates.
8-day weather	19 GeoTIFFs	01 Jun to 23 Oct 2025; 10 weather variables per period.
Static soil stack	24 bands	4 properties x 6 depths; 250 m, EPSG:32643.
Period manifest	19 rows	Master eight-day period calendar and new-script availability metadata.
Export tile grid	24 tiles	Reference grid for the new GEE export workflow.

Module 2: Real Satellite Temporal Table

A Python raster-sampling pipeline discovers all old files matching Karnal_ARD_YYYY-MM-DD_tileN.tif, transforms the 4,311 WGS84 observation locations into each raster CRS and samples whichever tile contains the point. The pipeline first inspects the band descriptions so that full and SAR-only periods can be handled without failure. Missing optical values remain NaN. Optical validity is derived from the availability of vegetation/moisture indices, while SAR validity requires real VV and VH values.

The resulting table contains 43,110 rows. Sentinel-1 is available for 100% of points on all ten dates. Optical validity varies substantially: 100% on 01 Jun, 31.18% on 17 Jun, 1.62% on 03 Jul, 0% on

19 Jul and 04 Aug, 25.68% on 20 Aug, approximately 100% on 05 Sep, 21 Sep and 07 Oct, and 41.89% on 23 Oct. This variability is retained as a feature of the real dataset rather than being hidden through unconditional interpolation.

Module 3: 19-Step Eight-Day Master Timeline

The master timeline uses the nineteen dates from the new weather manifest. For each eight-day step, the application retrieves fresh weather and independently searches the old satellite history for the most recent valid optical and SAR observations on or before the master date. When the source date equals the master date the modality is labelled FRESH; otherwise it is labelled CARRIED together with the age in days. If no earlier valid observation exists, it is UNAVAILABLE. Optical and SAR can therefore have different source dates at the same master step.

This design is particularly important on SAR-only satellite dates. For example, a date may use a carried NDVI/NDMI state from the last clear optical observation while simultaneously using a fresh VV/VH observation. The app displays these provenance states directly in the period bar and field inspector.

Module 4: Crop Engine

The Crop Engine compares Random Forest, XGBoost and RBF-SVM. The current tabular prototype uses EVI, NDMI, NDVI, NDWI, SAVI, VV, VH and the corrected VV - VH dB feature. Random Forest is selected for deployment because it is simple to serve, provides class probabilities and matched the best prototype Macro-F1. The current label space is Paddy (Rice) and Unclassified; this reflects the available derived labels and must not be interpreted as a complete Karnal crop inventory.

Module 5: Phenology Engine

The Phenology Engine compares Random Forest, XGBoost and a temporal LSTM. The tabular baselines operate on snapshot features, while the LSTM uses the real temporal table. Its input sequence contains NDVI, NDMI, NDWI, EVI, SAVI, VV, VH, VV_VH_ratio_dB, optical_valid, sar_valid and season_day. Unstable EVI values outside [-1,1] are converted to missing before imputation. Optical gaps are handled causally using field-wise forward fill only; future values are never back-filled into earlier dates.

The LSTM uses a three-period sequence with zero padding and a Masking layer for early-season observations. Entire field_id groups are separated across train, validation and test sets, resulting in 3,017 training fields, 647 validation fields and 647 held-out test fields. Because the current temporal table contains no independently observed phenology target, the notebook derives presentation-only growth_stage_pseudo labels from each field's real NDVI trajectory. The resulting confusion matrix therefore measures consistency with the pseudo-label generation process, not field-validated crop stage accuracy.

Module 6: Moisture Stress Engine

The Stress Engine compares Logistic Regression, Random Forest and XGBoost using the same corrected optical-SAR feature vector. XGBoost is selected for deployment in the current app. The present stress labels are derived and strongly imbalanced toward Severe Stress; consequently, Macro-F1 is reported alongside overall accuracy and the results are labelled prototype evaluation. Leakage-prone fields such as water_deficit_mm, recommended_depth_mm and canal_priority_score are excluded from model inputs.

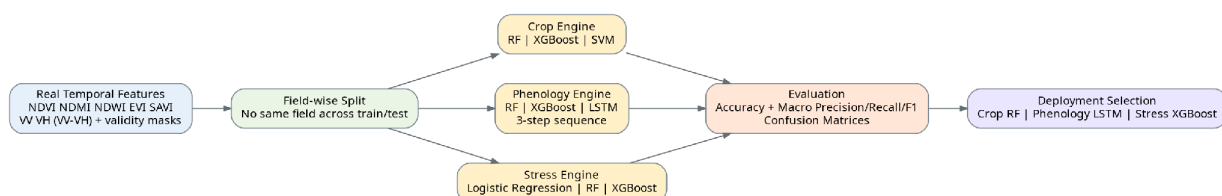


Figure 5.1: AI Engine Training and Evaluation Pipeline

Table 5.2: AI Engine Configuration

Engine	Models Compared	App Model	Primary Input	Current Target Status
Crop	Random Forest, XGBoost, SVM-RBF	RF	Snapshot optical + SAR features	Derived crop labels
Phenology	Random Forest, XGBoost, LSTM	LSTM	3-step real temporal sequence	Derived NDVI stage labels
Stress	Logistic Regression, Random Forest, XGBoost	XGBoost	Snapshot optical + SAR features	Derived stress labels
Irrigation	FAO-style water balance + rules	Rule engine	Weather + crop/stage + soil context	No confusion matrix yet

Module 7: Irrigation Engine

The irrigation component intentionally remains a transparent calculation rather than a classification model. The weather rasters provide period rainfall and the variables required for an FAO-56-style reference evapotranspiration estimate. Crop evapotranspiration is computed as $ET_c = K_c \times ET_0$. Prototype effective rainfall is estimated from period rainfall, and the water deficit is calculated as $Deficit = \max(0, ET_c - P_{eff})$. The recommended irrigation depth and priority class are then produced by threshold rules, with an optional escalation when the current real NDMI moisture signal is very low.

ERA5_PEV_mm is preserved and displayed as ERA5 potential evaporation; it is not relabelled as ET_0 . The app separately exposes the FAO-56 ET_0 estimate produced by the implemented meteorological calculation. Crop coefficients and effective-rainfall assumptions remain prototype values requiring local agronomic calibration before operational use.

Module 8: Role-Specific Web Application and Inference

The browser application uses Leaflet for the map and Chart.js for analytics. The Agriculture Officer sees the full regional interface and model-evaluation view. The Irrigation Officer and WUA roles are restricted to water-deficit and advisory layers. The Farmer role displays only two deterministically selected demo plots and hides regional model-development controls. The Policy Maker view emphasizes regional vegetation, deficit and priority summaries. These restrictions are user-interface behavior rather than secure production authorization and would require authenticated server-side role enforcement in a deployed multi-user system.

A persistent FastAPI inference service loads the selected Crop RF, Phenology LSTM, Stress XGBoost and their preprocessing artifacts once during service startup. Express proxies browser requests to this service. For a selected field and master date, the service reconstructs the latest independently valid optical and SAR snapshot for crop/stress prediction and builds the causal temporal sequence for the LSTM. The private_models directory is explicitly blocked from static web serving.

5.3 Data and API Schema

Table 5.3: API Endpoint Summary

Method	Endpoint	Description
GET	/api/data	Return master timeline, field metadata and base GeoJSON.
GET	/api/period/{date}	Return all field records for one eight-day master period.
GET	/api/field-timeline/{fieldId}	Return the 19-step history for one observation location.

Method	Endpoint	Description
GET	/api/model-evaluation	Return stored metrics and confusion-matrix arrays for the research view.
GET	/api/inference/health	Check whether Crop, Phenology and Stress artifacts are loaded.
GET	/api/inference/models	List selected deployment models and readiness.
GET	/api/inference/field/{fieldId}?as_of=YYY Y-MM-DD	Run live field inference at the selected master date.
POST	/api/run-pipeline	Rebuild master-period data from the local real datasets.

The main persistent files are karnal_real_temporal_table.csv for the ten-date real satellite sequence, karnal_master_demo_output.json for master metadata, nineteen data/periods/YYYY-MM-DD.json files for eight-day app state, model_evaluation.json for research metrics and selected serialized artifacts in private_models/.

CHAPTER 6

PROJECT IMPLEMENTATION & RESULTS

6.1 Introduction

This chapter reports what has actually been implemented and measured. Real data QA results are presented separately from prototype machine-learning results. This distinction is essential because the current crop, phenology and stress labels are derived or pseudo labels, whereas the satellite, weather and soil inputs are real downloaded datasets.

6.2 System Implementation

Backend and Inference Implementation

The current application uses Node/Express on port 3000 as the browser-facing server and FastAPI/Uvicorn on port 8001 as the persistent model-inference service. The Express process can automatically start the Python service. The Python registry loads the selected artifacts once, indexes the real temporal table by field_id and reuses the in-memory objects for later requests. This avoids repeatedly loading .joblib or .keras files and provides a clean boundary between the browser and Python ML runtime.

Frontend Implementation

The frontend uses Leaflet with satellite basemap tiles and Chart.js. The 19-step controller allows previous/next navigation, direct date selection, a range slider and automatic seasonal playback. Each master step shows whether the old satellite state is fresh, carried or unavailable, and the optical and SAR modalities can report different ages. Role selection changes the available layers, statistics, charts and field-inspection behavior without changing the underlying period data.

Application Screenshot Placeholders

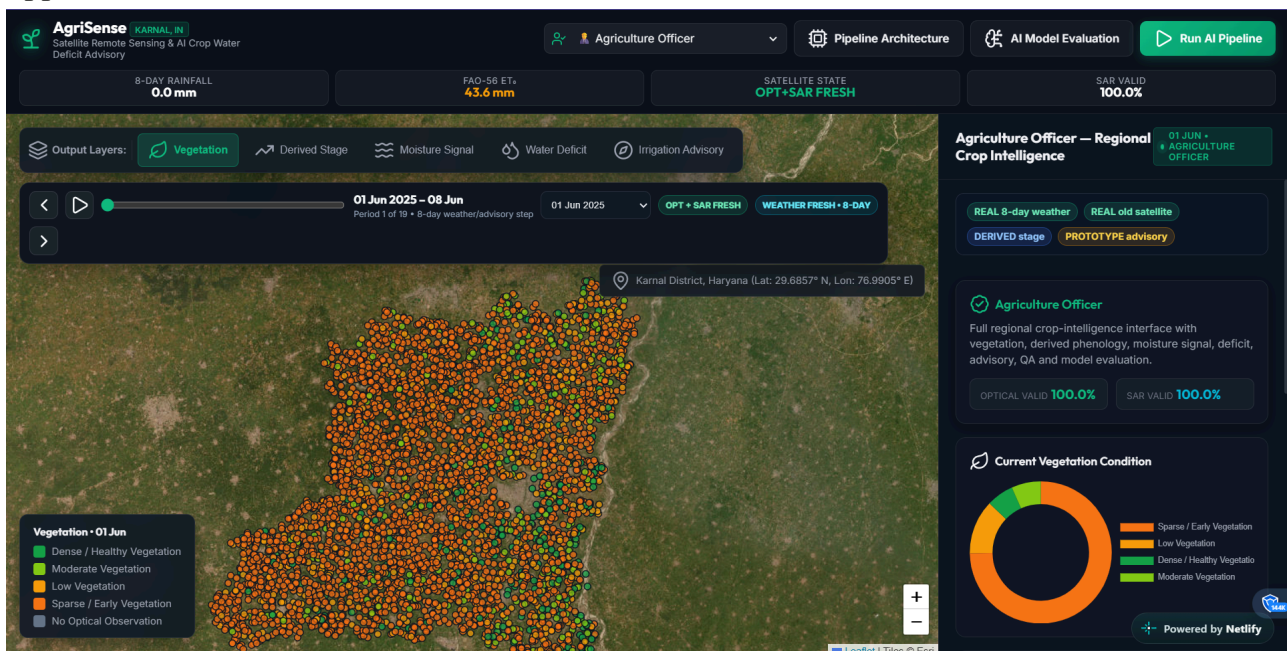


Figure 6.3: Agriculture Officer Dashboard Screenshot

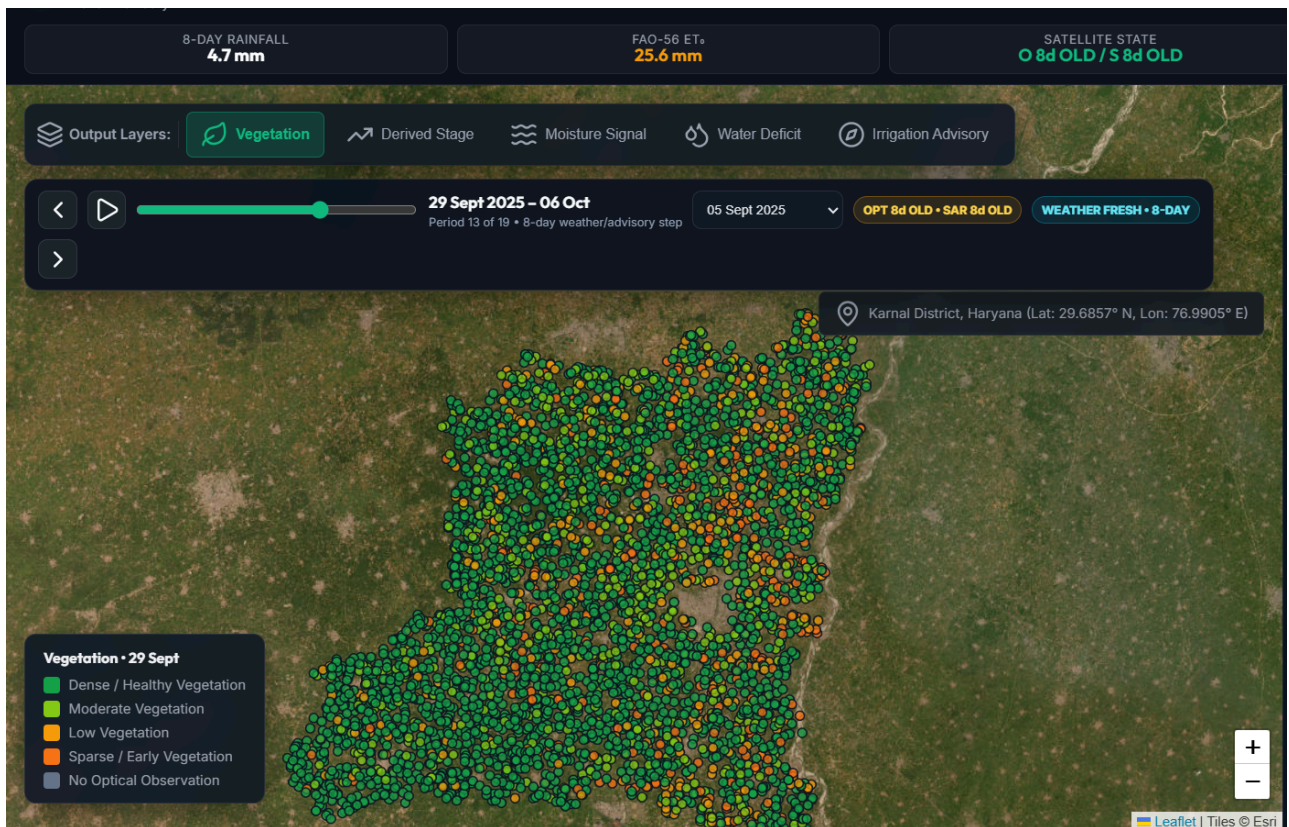


Figure 6.4: 19-Step 8-Day Master Timeline Screenshot

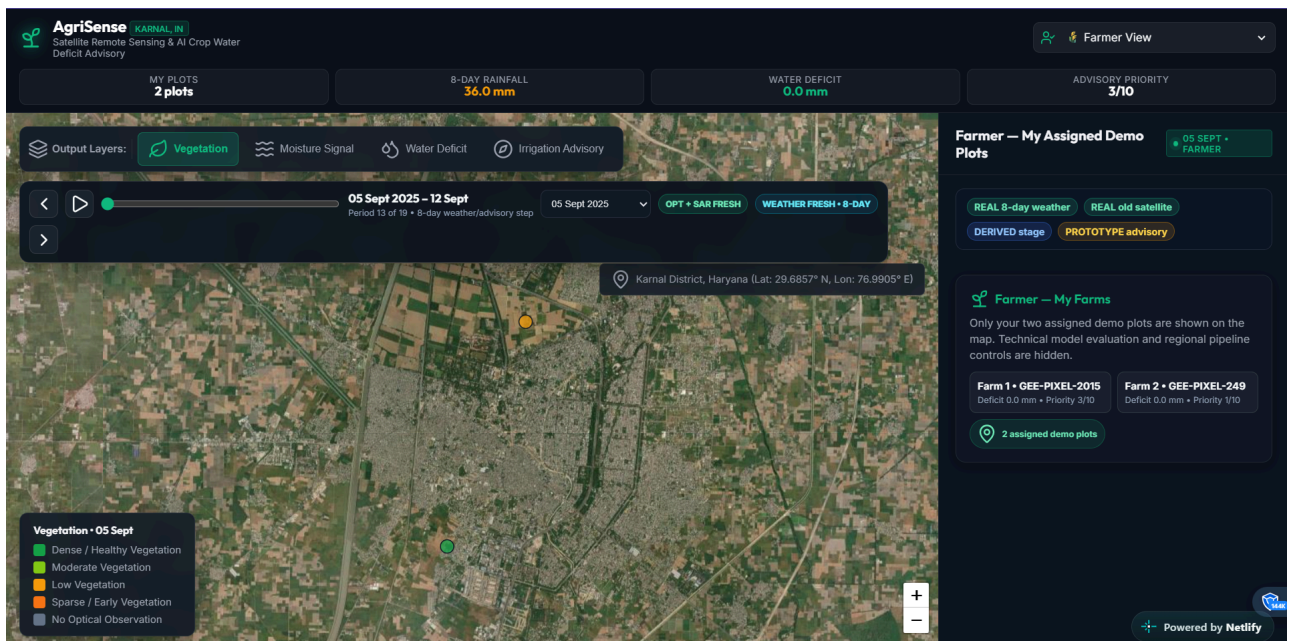


Figure 6.5: Farmer Role-Specific Interface Screenshot

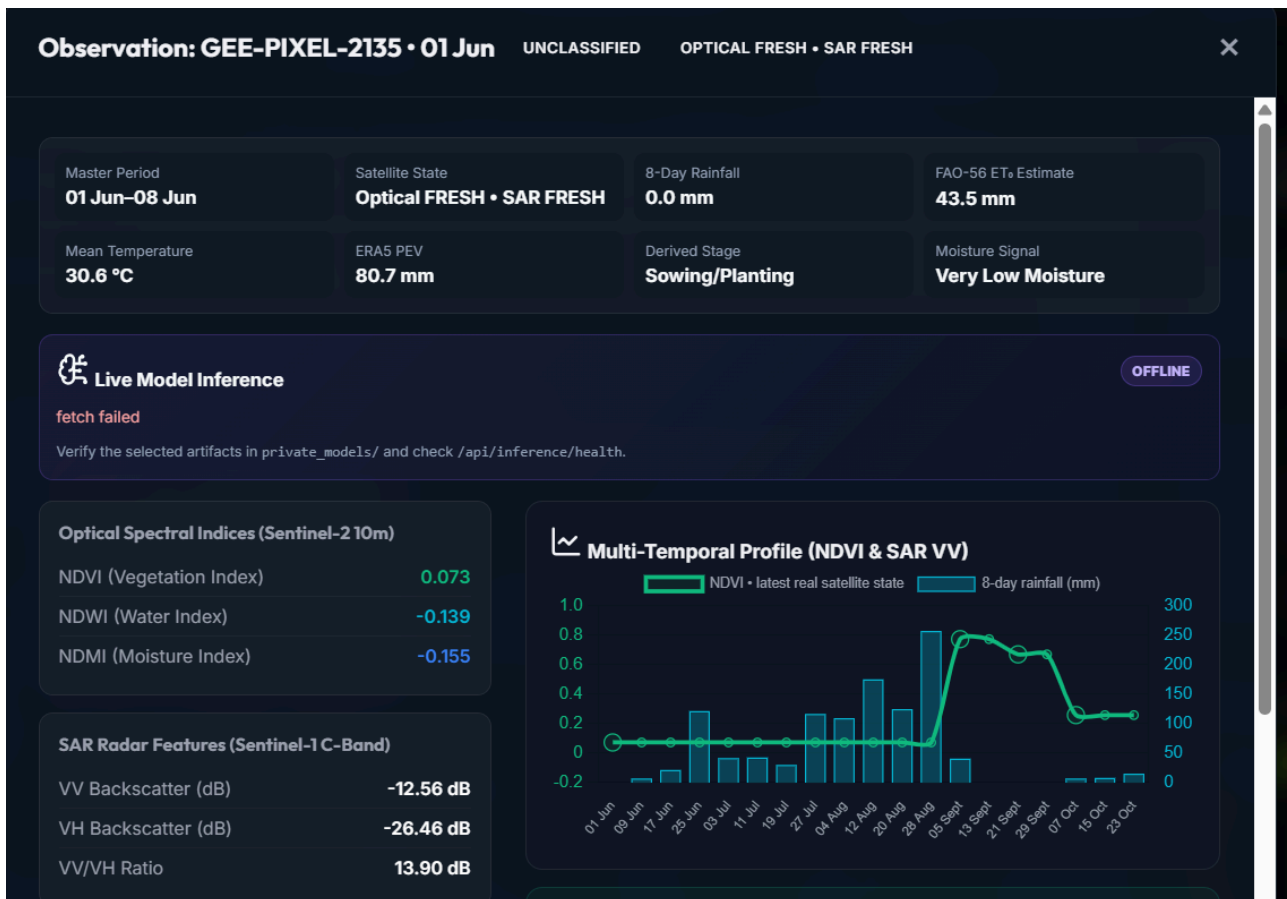


Figure 6.6: Field Inspector with Live Model Inference Screenshot

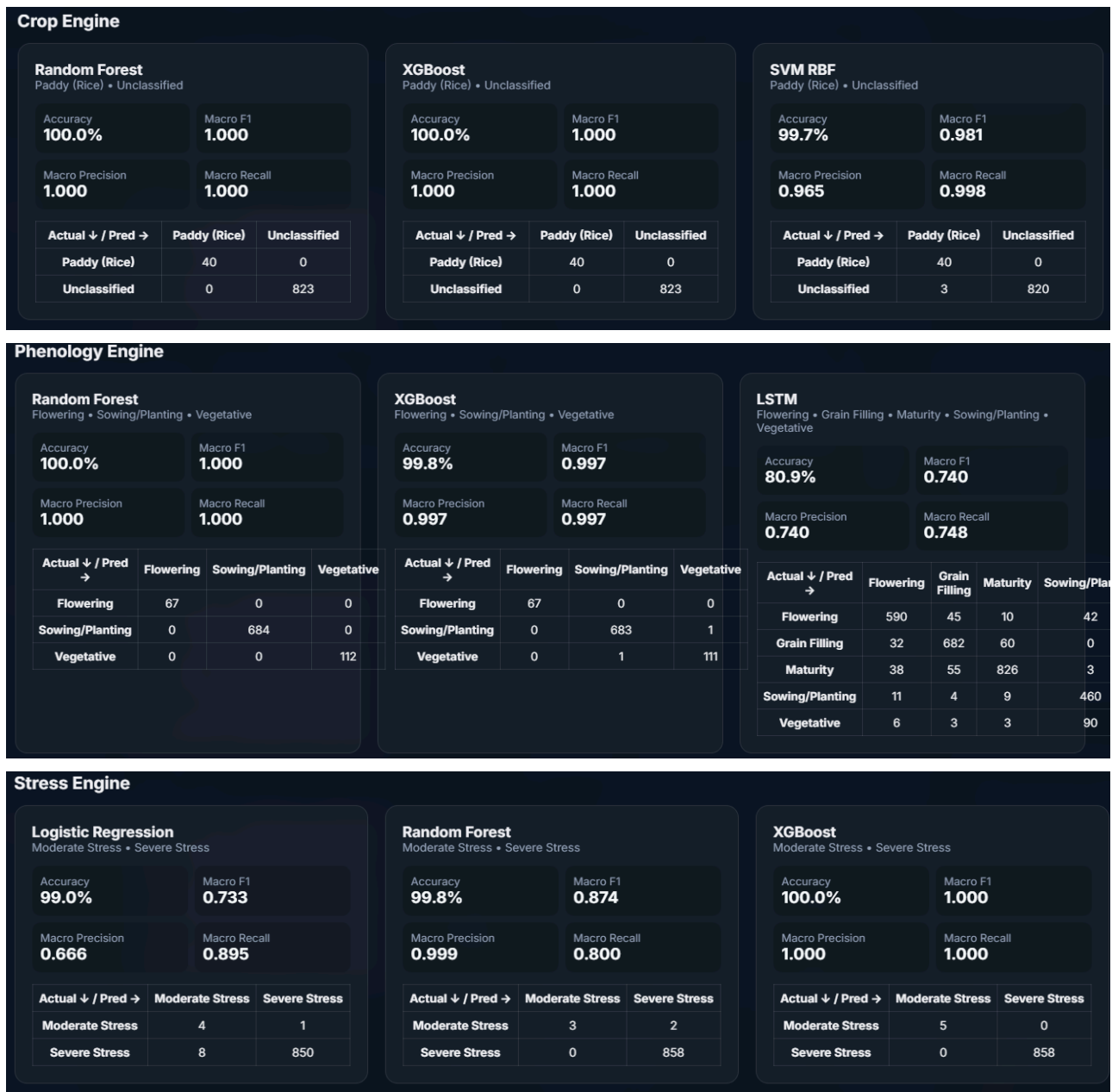


Figure 6.7: AI Model Evaluation Screen Screenshot

6.3 Performance Metrics & Results

Table 6.1: Real Dataset and Temporal QA Summary

Metric	Achieved Result	Interpretation
Old satellite files	110 GeoTIFFs	88 full 14-band + 22 SAR-only 3-band files.
Satellite periods	10 dates	01 Jun 2025 to 23 Oct 2025.
Fixed locations	4,311	Same locations sampled through the old satellite season.
Temporal table	43,110 rows	4,311 x 10 field-period records.
SAR coverage	100% on all 10 dates	VV and VH available at every sampled location/date.
Optical coverage	0-100% depending on date	Cloud/scene limitations preserved explicitly.
Weather timeline	19 periods	Complete eight-day sequence from 01 Jun to 23 Oct 2025.

Metric	Achieved Result	Interpretation
Weather raster	10 bands, 5.5 km	Rainfall, temperature, dew point, wind, pressure, radiation, PEV, validity.
Soil stack	24 bands, 250 m	Texture, bulk density, organic carbon and field capacity at 6 depths.

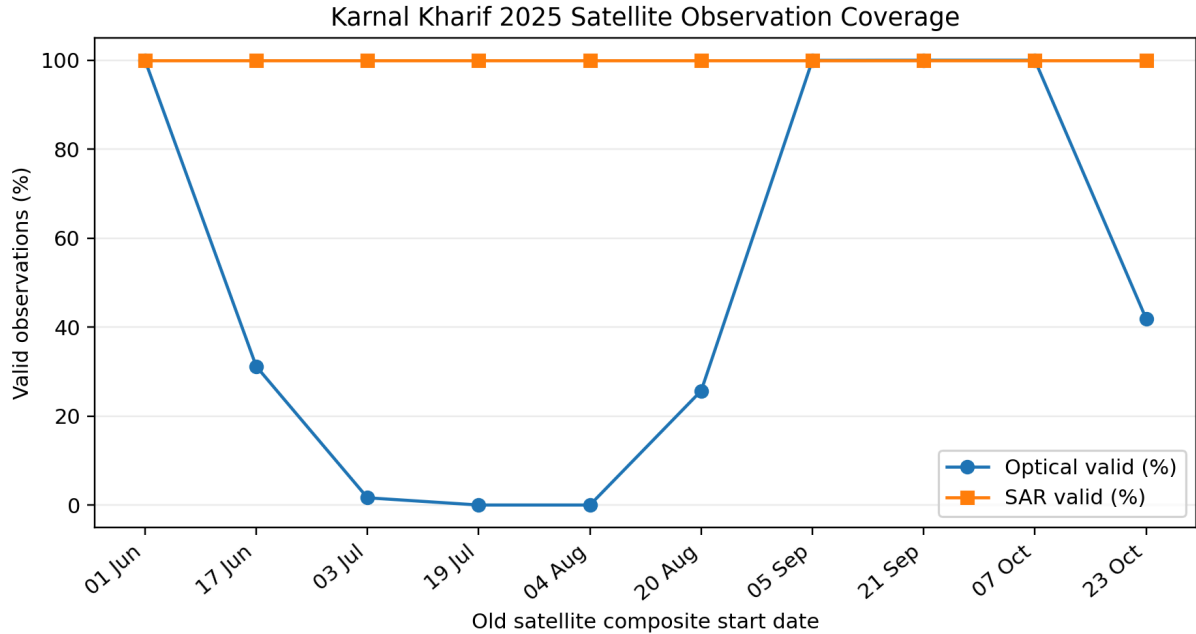


Figure 6.1: Karnal Kharif 2025 Satellite Observation Coverage

The coverage figure demonstrates the value of the optical-SAR fusion strategy. Optical validity falls to 1.62% on 03 July and 0% on 19 July and 04 August, while SAR remains at 100%. The application therefore carries the most recent valid optical state when necessary but can still use fresh SAR observations. This is a real data-availability pattern rather than a simulated robustness test.

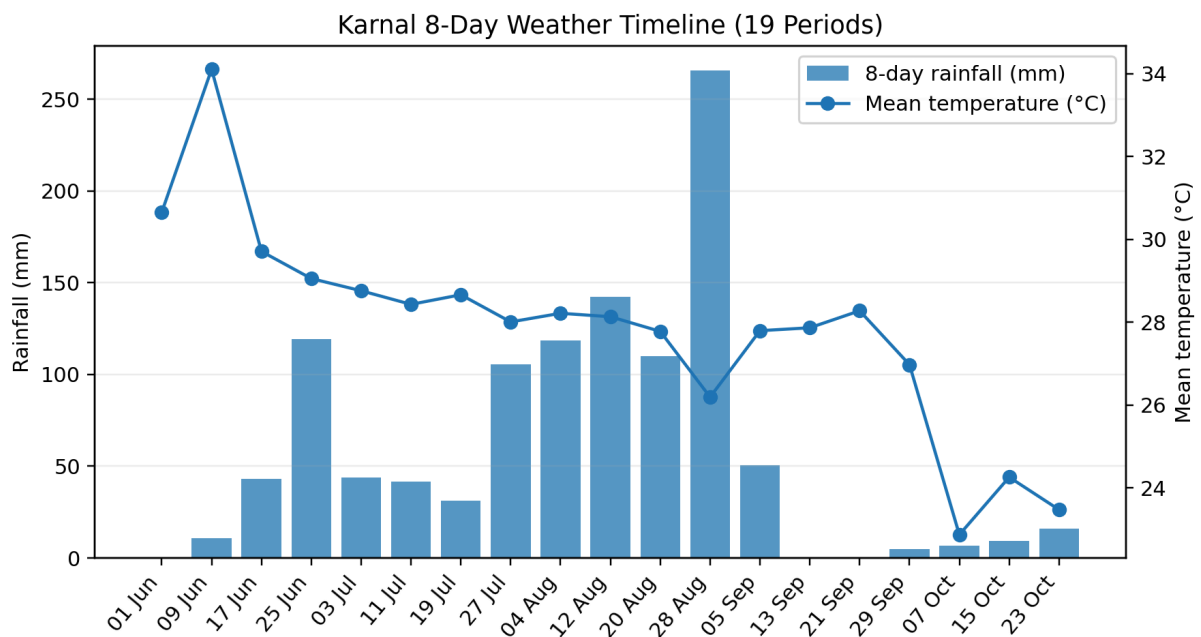


Figure 6.2: Karnal 8-Day Weather Timeline

The weather sequence is complete for all nineteen master periods. Period-mean rainfall varies from 0 mm to approximately 265.75 mm across Karnal, while mean period temperature ranges from approximately 22.87 C to 34.10 C. The late-August high-rainfall period and the cooler October conditions provide useful seasonal context for the irrigation rule engine.

Table 6.2: Prototype AI Model Comparison

Engine	Model	Accuracy	Macro Precision	Macro Recall	Macro F1
Crop	Random Forest	100.00%	1.000	1.000	1.000
Crop	XGBoost	100.00%	1.000	1.000	1.000
Crop	SVM RBF	99.65%	0.965	0.998	0.981
Phenology	Random Forest	100.00%	1.000	1.000	1.000
Phenology	XGBoost	99.77%	0.997	0.997	0.997
Phenology	LSTM	80.89%	0.740	0.748	0.740
Stress	Logistic Regression	98.96%	0.666	0.895	0.733
Stress	Random Forest	99.77%	0.999	0.800	0.874
Stress	XGBoost	100.00%	1.000	1.000	1.000

The numerical values in Table 6.2 are prototype evaluation results on derived or pseudo labels. The perfect or near-perfect tree-model scores should therefore not be interpreted as independent agronomic accuracy. In particular, the current Crop Engine is a two-class Paddy/Unclassified problem and the current stress labels are heavily imbalanced. Macro-F1 is included to make minority-class behavior more visible.

Table 6.3: Selected Deployment Models and Interpretation

Engine	Selected Model	Accuracy / Macro F1	Interpretation
Crop Engine	Random Forest	1.000 / 1.000	Selected for simple probability-based tabular deployment; current labels are Paddy vs Unclassified.
Phenology Engine	LSTM	0.809 / 0.740	Selected because it actually uses the real temporal sequence; five pseudo-stage classes.
Stress Engine	XGBoost	1.000 / 1.000	Selected prototype stress model; results depend on derived, imbalanced labels.
Irrigation Engine	FAO-style rule engine	N/A	No confusion matrix until observed irrigation/water-status outcomes are available.

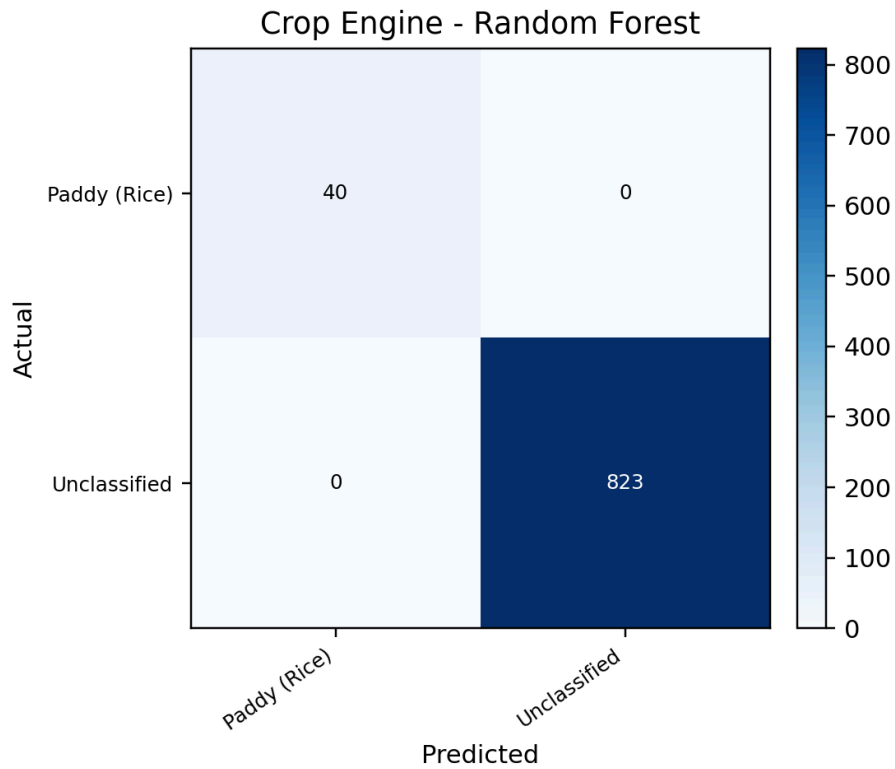


Figure 6.8: Crop Engine Random Forest Confusion Matrix

The Crop Random Forest confusion matrix contains 40 Paddy observations and 823 Unclassified observations in the held-out prototype evaluation with no observed errors. Because the labels are derived and highly imbalanced, this matrix primarily verifies implementation consistency and cannot establish real crop-mapping accuracy.

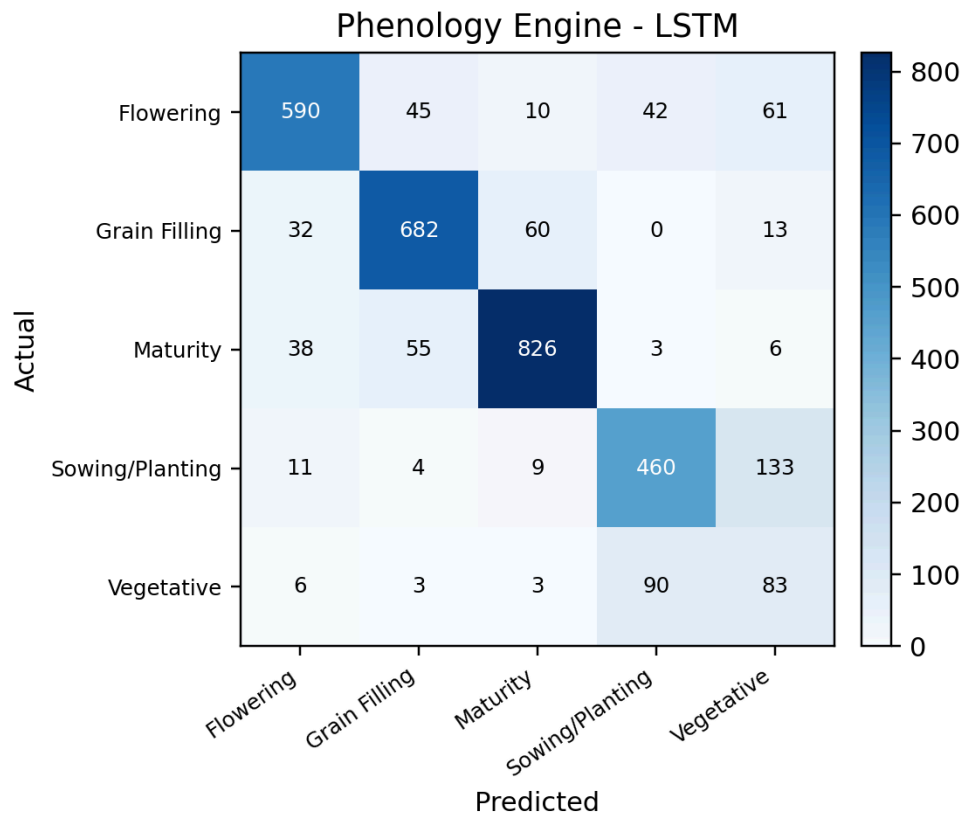


Figure 6.9: Phenology Engine LSTM Confusion Matrix

The temporal LSTM evaluates five pseudo-stage classes and achieves 80.89% accuracy, 0.740 Macro-F1 and 0.817 weighted F1. Its errors occur mainly among neighboring stage categories, including confusion between Sowing/Planting and Vegetative and between Flowering and later stages. The model is evaluated on held-out field groups, but the target itself is still derived from the NDVI trajectory.

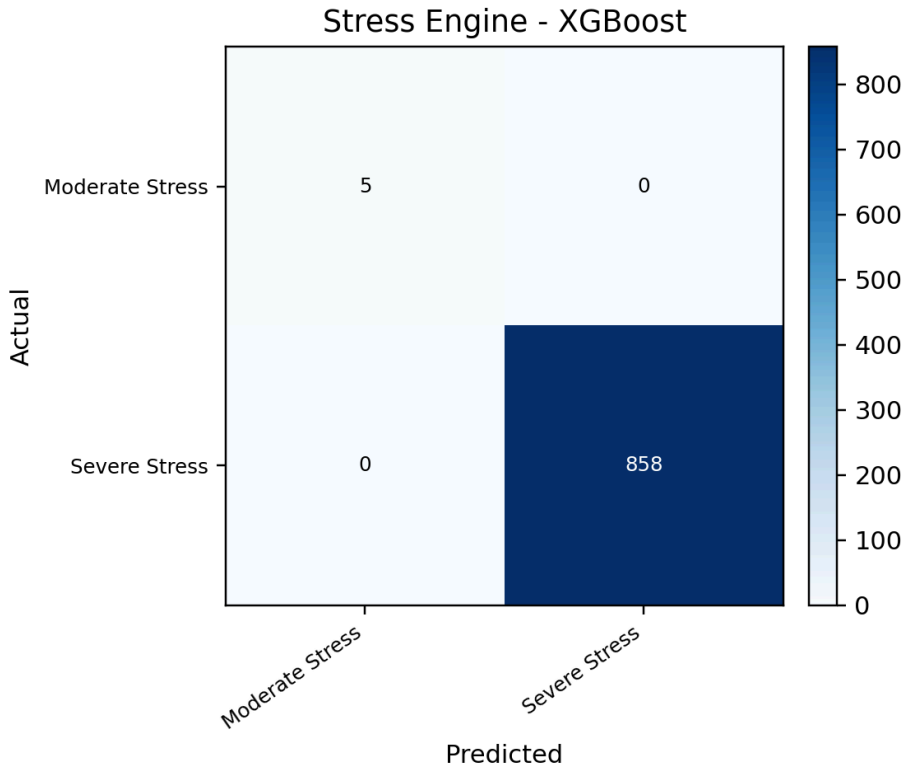


Figure 6.10: Stress Engine XGBoost Confusion Matrix

The Stress XGBoost prototype classifies 5 Moderate Stress and 858 Severe Stress held-out samples without error in the current derived-label test set. The extreme class imbalance and pseudo-label source make independent field validation essential before the model can be used for agronomic decisions.

Current Result Interpretation

The strongest result of the current project is the complete end-to-end engineering chain: real multi-date satellite sampling, explicit cloud-gap handling, complete eight-day weather integration, multi-depth soil context, a 19-step master timeline, multiple ML engines, role-specific visualization and persistent live inference. The present model metrics demonstrate that the software can train, serialize, evaluate and serve the selected models, but they do not replace independent crop, phenology or moisture-stress ground truth.

CHAPTER 7

CONCLUSION & FUTURE SCOPE

7.1 Conclusion

AgriSense converts the Karnal 2025 Kharif problem statement into a working multi-source agricultural intelligence prototype. The project successfully reused a complete old satellite download rather than discarding it after the newer eight-day satellite export was constrained by GEE quota. A variable-band-aware local pipeline converted 110 GeoTIFF tiles into a 43,110-row temporal table while preserving periods with partial or missing optical coverage. Sentinel-1 remained fully available across the ten old satellite dates and therefore provides a practical backstop during cloudy monsoon periods.

The complete new eight-day weather sequence, 24-band soil stack, period manifest and tile grid were integrated with the old satellite data to construct a 19-step master seasonal timeline. The application does not invent satellite imagery on intermediate dates. Instead, optical and SAR observations are independently labelled fresh, carried or unavailable and their source date and age are exposed to the user.

Three AI engines and one transparent irrigation engine have been implemented. Random Forest, XGBoost and SVM are compared for crop classification; Random Forest, XGBoost and LSTM are compared for phenology; and Logistic Regression, Random Forest and XGBoost are compared for moisture stress. The current deployed combination is Crop Random Forest, temporal Phenology LSTM and Stress XGBoost. These artifacts are loaded once by a persistent FastAPI service and accessed through the Express server from the role-specific web app.

The project also demonstrates an important limitation of prototype AI evaluation. The current crop, stage and stress targets are derived or pseudo labels. Very high confusion-matrix scores on such targets verify consistency with the generated labels, not independent agronomic validity. The report therefore keeps these results explicitly separated from the real satellite, weather and soil QA results.

Key Outcomes of the Project:

- A real Karnal satellite temporal table with 4,311 locations, 10 dates and 43,110 records.
- Complete 19-period eight-day weather coverage and a 24-band multi-depth soil stack.
- Independent optical and SAR fresh/carried logic for the eight-day master timeline.
- Three evaluated AI engines with serialized selected deployment models.
- A temporal three-step LSTM phenology pipeline using causal gap handling and field-group splits.
- A transparent FAO-style water-balance and irrigation-priority engine.
- A role-specific Leaflet/Chart.js application with seasonal playback, field inspection and model-evaluation views.
- A persistent FastAPI inference layer integrated with the Express web server without exposing private model files.

7.2 Future Scope

Short-Term Enhancements (1-3 Months):

- Replace demo/pseudo crop, phenology and stress targets with independently collected field labels from Karnal.
- Insert the final application screenshots into the placeholders in Chapter 6 and complete UI usability testing.
- Calibrate crop coefficients, effective-rainfall assumptions and irrigation-depth thresholds using local agronomic guidance.
- Add secure authentication and server-side role authorization so farmer and officer data scopes are enforced beyond the interface.

- Add automated data QA warnings for stale optical observations, missing weather cells and abnormal index values.

Medium-Term Enhancements (3-6 Months):

- Integrate verified field boundaries and real WUA/canal-command polygons instead of using point observations and regional demo scopes.
- Add independent soil-moisture, irrigation-event or crop-water-status observations for stress and irrigation validation.
- Evaluate ascending and descending SAR streams separately and compare improved speckle/terrain-normalization methods.
- Expand phenology modelling with Temporal CNN or transformer encoders once adequate labelled sequences are available.
- Deploy the app with a spatial database and optimized raster/COG delivery for larger user populations.

Long-Term Enhancements (6-12 Months):

- Validate across multiple Kharif/Rabi seasons and additional districts before claiming geographic generalization.
- Integrate canal-release data, farmer irrigation logs and water-availability constraints into the advisory engine.
- Develop multilingual mobile access, offline field inspection and push notifications for actionable irrigation alerts.
- Introduce calibrated probabilistic uncertainty and explainability showing which dates and features contributed to each model output.
- Extend the platform from a project prototype into an operational decision-support system only after independent agronomic validation.

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